ASSEMBLY DYNAMICS FOR A CORAL-ASSOCIATED REEF COMMUNITY:

SPATIAL AND TEMPORAL PATTERNS, AND A TEST OF PRIORITY EFFECTS

A DISSERTATION SUBMITTED TO THE GRADUATE DIVISION OF THE UNIVERSITY OF HAWAIʻI AT MĀNOA IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF

DOCTOR OF PHILOSOPHY

IN

MARINE BIOLOGY

DECEMBER 2018

By

Chelsie W. W. Counsell

Dissertation Committee:

Megan J. Donahue, Chairperson
Erik C. Franklin
Mark A. Hixon
Margaret A. McManus
Amber N. Wright

Keywords: community ecology, cryptofauna, *Pocillopora meandrina*, Hawaiʻi

PREVIEW

ProQuest Number: 13421246

All rights reserved

INFORMATION TO ALL USERS
The quality of this reproduction is dependent upon the quality of the copy submitted.

In the unlikely event that the author did not send a complete manuscript and there are missing pages, these will be noted. Also, if material had to be removed, a note will indicate the deletion.

ProQuest

ProQuest 13421246

Published by ProQuest LLC (2019). Copyright of the Dissertation is held by the Author.

All rights reserved.
This work is protected against unauthorized copying under Title 17, United States Code
Microform Edition © ProQuest LLC.

ProQuest LLC.
789 East Eisenhower Parkway
P.O. Box 1346
Ann Arbor, MI 48106 – 1346

PREVIEW

© Copyright 2018 – Chelsie Counsell

All Rights Reserved.

PREVIEW

DEDICATION

This work is dedicated to the memory of Dr. Susan Williams for sharing her passion for marine ecology with myself and many others, encouraging women to pursue careers in science, and inspiring us to share science broadly with colleagues, politicians, and community members.

PREVIEW

ACKNOWLEDGMENTS

I would like to thank my committee, Megan Donahue, Mark Hixon, Margaret McManus, Erik Franklin, and Amber Wright for their support and assistance throughout my graduate career. Discussions with my committee members have helped me formulate my approach to designing, implementing, and presenting ecological research, and I look forward to continuing to work with each of them. In particular, I would like to thank my advisor Megan Donahue for our numerous conversations regarding the design, analysis, and writing of my dissertation research. Further, Megan has strengthened my science writing, has helped me stay excited about science through the entire research process, has encouraged me to explore various skill-building opportunities, and has helped me push through roadblocks along my dissertation journey. I would also like to acknowledge the mentors who guided me through my early research experiences and encouraged me to pursue a Ph.D.: Michael Kingston was my undergraduate advisor at Elon University and along with John Claydon at the Center for Marine Resource Management Studies, mentored my undergraduate honors thesis. Eric Sanford was my mentor for a summer research experience for undergraduates (REU) and Susan Williams was the director of this REU program at the Bodega Bay Marine Laboratory. Felicia Coleman and Kevin Craig advised my Master's thesis at Florida State University.

The Donahue Lab has continually been a source of life advice, dive buddies, field gear, intellectual inspiration, and friendship, and I am grateful to be a part of this incredible group of scientists. I would like to thank the graduate student community at the Hawai‘i Institute of Marine Biology (HIMB) and the University of Hawai‘i (UH) for their ongoing camaraderie and support. I would especially like to thank my colleagues who agreed to collaborate with me on side research projects while also working on their dissertations: Richard Coleman, Erika Johnston, and Tayler Massey. I would also like to thank the HIMB community for welcoming me into their ‘ohana, particularly the boat drivers and grounds crew who always go out of their way to brighten everyone's day with smiles, helping hands, and even ukulele jam sessions.

My Ph.D. was supported with a National Science Foundation (NSF) Graduate Research Fellowship (Grant No. 2012103208). Additional funding was provided by a Colonel Willys E. Lord DVM and Sandina L. Lord Endowed Scholarship, an American Academy of Underwater Sciences (AAUS) Foundation Kathy Johnston English Scholarship, a Marine Biology Program

PREVIEW

Small Grant, the Donahue Lab, and the Castle Foundation (Grant 3846, M. Hixon PI). Logistical support for boat and dive operations was provided by the Dive Safety Offices (DSO) at HIMB and UH Mānoa. Jeff Kuwabara, Garrett Johnston, and Jason Jones helped coxswain small boats. I would particularly like to thank Jason Jones for the many conversations on science diving logistics which ensured my research goals were met while prioritizing diver safety, for getting in the water on multiple occasions to support my research team, and for working to maintain the dive and boat operations at HIMB. I am also grateful to Maria Laamang and the support staff at UH's DSO for the various behind-the-scenes efforts that keep UH's diving operations running smoothly. Additional fieldwork was supported by Maui Community College, UH Hilo, and the National Oceanic and Atmospheric Administration (NOAA).

Many people have shared countless hours of their time to help me complete my dissertation fieldwork: Karen Bryan, Jamie Caldwell, Iain Caldwell, Kim Cobb, Richard Coleman, Courtney Couch, Rebecca Counsell, Travis Counsell, Megan Donahue, Erik Franklin, Tara Hetz, Mark Hixon, Conor Jerolmon, Colton Johnson, Ingrid Knapp, Claire Lager, Katie Lubarsky, Amy Moran, Kailey Pascoe, Miranda Rancourt, Megan Ross, Megan Russell, Rebecca Weible, and Julie Zill. I have also had the privilege of working with a team of highly devoted research interns: Tiffany-Nicole Gutlay, Tate Wester, Julia Stewart, Clara Schlieman, and Sarah Franklin. I'd like to particularly thank four colleagues who spent exceptionally large amounts of time in the field with me: Erika Johnston, Debra Ford, Eileen Nalley, and Nyssa Silbiger. Without the support of all these people, my dissertation would not have been possible, I would have forgone many midday meal breaks, and I would have quit after the second broken bone. Thank you for going above and beyond dive buddies.

Finally, I am forever grateful to my friends and family for their ongoing support. This includes Stellaluna and Rufus, who have been a continual source of emotional support and an untiring audience for practice talks. This also includes my life partner, Travis Counsell, who has always believed in me and supported my career aspirations through the especially challenging days, as well as the good ones.

PREVIEW

ABSTRACT

A foundational goal of community ecology is to understand the drivers of variation in communities over spatial and temporal scales. This dissertation combines nondestructive, observational studies of natural variation within a coral-associated reef community over spatial and temporal scales, with a community manipulation that directly tested the effects of two common species on community assembly processes. The observational components of this research characterized the composition of coral-associated reef communities and investigated patterns in community composition relative to colony- and regional-scale environmental gradients. The results showed that colony-scale parameters influence the composition of these communities more than regional-scale parameters over both spatial and temporal scales. Colony size, depth, and percent live coral tissue were all correlated with changes in community composition in space and time. In addition to patterns associated with colony-scale parameters, communities varied along spatial gradients in wave height and chlorophyll-a. The composition of communities was more consistent in time than in space when comparing within-colony and between colony community dissimilarity metrics. Spatial surveys depicted non-random co-occurrence patterns between some of the commonly observed community members. By modeling probabilities of species arrival and departure relative to the presence of taxonomically similar species, temporal surveys of focal communities reinforced the importance of species interactions in structuring coral-associated reef communities and added a directional component to these potential species interactions. Community manipulations directly quantified the effects of two commonly observed protection mutualist species on the formation and maintenance components of community assembly. The results of these experiments indicated that these two species have inhibitory effects on the composition of the coral-associated community. These

PREVIEW

effects were driven by species-specific responses to each of the mutualist species. This dissertation provides fundamental details about the natural variation in coral-associated communities, incorporates a community manipulation to directly test community assembly processes, and provides a template of analyses for characterizing spatial and temporal patterns in other communities.

PREVIEW

TABLE OF CONTENTS

Acknowledgments iii
Abstract v
List of Tables ix
List of Figures xiii
Chapter 1 1
Introduction 1
Community assembly 1
Study system 3
Research goals 4
Chapter 2 6
Variation in coral-associated cryptofaunal communities across spatial scales and environmental gradients 6
Abstract 6
Introduction 7
Methods 10
Results 13
Discussion 16
Chapter 3 45
Species sorting drives temporal variation in the composition of coral-associated metacommunities 45
Abstract 45
Introduction 46
Methods 50
Results 55
Discussion 57
Chapter 4 75
Priority effects and community assembly: effects of protection mutualists on coral reef cryptofauna 75
Abstract 75

PREVIEW

Introduction 76
Methods 79
Results 84
Discussion 87
Chapter 5 107
Conclusions 107
Main findings 107
Future directions 108
Research significance 110
Literature Cited 111

PREVIEW

LIST OF TABLES

Table 2.1. Environmental driver variables, including the source of data, description of variable, unit, mean value, and range of values. Means and ranges are calculated for all 751 *Pocillopora* colonies surveyed, except for inter-branch distance, which was measured for only 716 colonies, and density of *Pocillopora*, which was available for 743 colonies…….. 23

Table 2.2. Family, genus, species, species codes, % of regions, % of sites, % of colonies, average number of individuals observed on a colony, depth range, and % live coral tissue range for fish (n=10) and invertebrate (n=21) species observed on ≥1% of surveyed *Pocillopora meandrina* (n=751 colonies)……...…….. 24

Table 2.S1. Correlation coefficients between environmental drivers measured at the colony and site scale; scores larger than ±0.5 listed in bold. Correlation scores for inter-branch distance and *Pocillopora* density are based on a subset of 708 colonies for which measurements of these variables were available. All other scores are based on 751 colonies…….. 31

Table 2.S2. Marginal R^2 (variance explained by fixed effects) and conditional R^2 (variance explained by full model including fixed and random effects) for a set of binomial GLMMs run with probability of occurrence for the species observed on ≥5% of colonies as the response variable…….. 32

Table 2.S3. Family, genus, species, % of regions, % of sites, % of colonies, average number of individuals observed on a colony, depth range, and % live coral tissue range listed for all fish (n=48) and invertebrate (n=43) species observed on 751 *Pocillopora meandrina*…….. 33

Table 2.S4. List of site names, abbreviations, coordinates, regions, and depth ranges for 19 survey sites around O‘ahu, Hawai‘i……...…….…….. 36

Table 3.1. Environmental driver variables, including the source of data, description of metric included in analyses, and type of parameter……...……...62

Table 3.2. Partial Mantel statistic R^2s are presented for a set of tests run with Bray-Curtis community based distances as the response matrix and an environmental matrix of colony characteristics, biophysical parameters, or both (Table 3.1) as the explanatory matrix. All comparisons were conditional on temporal distance. For both sites, most variation in community

PREVIEW

composition was explained by colony characteristics. Mantel statistics that are significant at p <0.05 are in listed in bold..……....63

Table 3.3. Effect sizes (parameter estimates) for biophysical parameter and colony characteristic predictor variables (Table 3.1) from regression models on community level diversity metrics (bold if significant at p <0.05). Prior to analyses all predictor variables were centered and scaled, and response variables were detrended through time.……..…….....64

Table 3.4. Effect sizes (parameter estimates) for the presence of taxonomically similar species at time *t* on the probability of arrival (top) or persistence (middle) for each of five trapeziid species at time *t+1* from generalized linear mixed models. Effect sizes significant at p <0.1 are colored red for negative effects and blue for positive effects. Effect sizes significant at p <0.05 are in bold. The bottom table displays non-random spatial co-occurrences (negative in red, positive in blue; spatial co-occurrence between $species_a$ and $species_b$ is the same as between $species_b$ and $species_a$) from a survey of 751 colonies across 19 different sites (Counsell et al. 2018)……..65

Table 3.5. Effect sizes (parameter estimates) for the presence of taxonomically similar species at time *t* on the probability of arrival (top) or persistence (middle) for each of three Scorpaenidae species at time *t+1* from generalized linear mixed models. Effect sizes significant at p <0.1 are colored red for negative effects and blue for positive effects. Effect sizes significant at p <0.05 are in bold. *S*. complex refers to a species complex of three visually similar scorpionfish species (*Sebastapistes fowleri*, *S. galactacma*, and *S. ballieui*) that were not consistently distinguished to species. The bottom table displays non-random spatial co-occurrences (negative in red, positive in blue; spatial co-occurrence between $species_a$ and $species_b$ is the same as between $species_b$ and $species_a$) from a survey of 751 colonies across 19 different sites (Counsell et al. 2018)…...…...66

Table 3.S1. Family, genus, species, and relation to host coral based on literature (i.e., mutualist with known benefits, parasite with known negative effects, or commensal with unknown effects) for species observed during repeated surveys (>3 years) on focal colonies (n=42). Species are listed in descending order relative to their persistence on individual colonies (the probability that they were observed on the same colony for two consecutive time steps). Only includes the species that were more persistent than predicted for their observed spatial occupancy (proportion of colonies occupied, averaged across time steps) based on a null model with random distributions across colonies at each timestep (Fig. 3.2).…….....70

Table 3.S2. Family, genus, species, and relation to host coral based on literature (i.e., mutualist with known benefits, parasite with known negative effects, or commensal with unknown effects)

PREVIEW

for species observed during repeated surveys (>3 years) on focal colonies (n=42). Species are listed in descending order relative to their persistence on individual colonies (the probability that they were observed on the same colony for two consecutive time steps). Only includes the species whose persistence was similar to or less than that predicted for their observed spatial occupancy (proportion of colonies occupied, averaged across time steps) based on a null model with random distributions across colonies at each timestep (Fig. 3.2).……71

Table 3.S3. AIC scores from a set of models describing how within-colony community dissimilarity varies across a temporal gradient: $dissimilarity\ (\Delta t) = \Phi_1 + (\Phi_2 - \Phi_1) \times e^{-\Phi_3 \times \Delta t}$. In all models, Φ_1 is the asymptote of predicted maximum within-colony dissimilarity reached after long periods of time between surveys; Φ_2 is the intercept for within-colony dissimilarity predicted when there is no time between surveys; and Φ_3 is the rate of increase for within-colony dissimilarity as the amount of time between surveys increases. For each model, the parameters that were allowed to vary by coral colony nested in site are in bold. For some models, the intercept Φ_2 was set to zero: $dissimilarity\ (\Delta t) = \Phi_1 \times (1 - e^{-\Phi_3 \times \Delta t})$ ...……..72

Table 4.1. Model output from binomial GLMs for the proportion of sampling days that colonizers were observed on corals. Treatment output (function "Anova" in *R* package *car*; Fox et al. 2018) and size output (function "summary" in *R* package *base*; Chambers and Hastie 1992) from models run with these two variables as fixed effects and grid row (i.e., treatment block) as a random effect……..……...93

Table 4.2. Model output from GLMMs for community members on corals during a community establishment experiment. Output (function "Anova" in *R* package *car*; Fox et al. 2018) from models run with treatment (df=3), sampling time (the models do not include week zero as this week was defined by the experiment), and an interaction term (df=3) as fixed effects, and colony individual as a random effect.…….....94

Table 4.3. Model output for community level parameters relative to the decapod communities on corals manipulated (n=40) to have a pair of *A. lottini* or *Tr. intermedia*, both, or neither and then followed for six months (surveyed eleven times), and on naturally occurring corals surveyed around O'ahu (n=321, surveyed once). All models had the community level parameter as the response variable, treatment as a fixed effect, and for the models on experimental communities time and the interaction of time and treatment were also included as fixed effects……..……....95

PREVIEW

Table 4.S1. Family, genus, species, classification as mutualist (mu), commensal (co), or facultative (fa) relative to *Pocillopora meandrina* (based on existing literature, e.g., Coles 1980, Black and Prince 1983, Stella et al. 2011, Britayev et al. 2017), and summaries of occurrence and frequency for all mobile decapod species observed across surveys of a colonization experiment, an establishment experiment, and naturally occurring communities around O‘ahu (n=321).….101

PREVIEW

LIST OF FIGURES

Figure 2.1. Location and name codes of 19 study sites around O‘ahu. For each site the number of colonies surveyed is given in parentheses. Sites are grouped by color into regions. Inset map of the main Hawaiian Islands with O‘ahu outlined. See Table 2.S4 for site names and coordinates..... 25

Figure 2.2. Characterization of community. (a) Species rarefaction curve for the overall pool of species found associated with 751 *Pocillopora meandrina*. The dashed line shows the mean Chao estimate for number of species, which reaches 115 species for 751 colonies. (b) For each of 91 species, the mean number of individuals observed and the variance in the number of individuals observed across colonies display a relative index of aggregation, where species that fall above the 1:1 line have clumped distributions. (c) Species rank abundance plot with decapods species shown in red, other invertebrates in grey, and fishes in blue..... 26

Figure 2.3. Multi-model-averaged parameter coefficients with 95% confidence intervals from the subset of models with ΔAIC <4 for each community metric, i.e., species richness (black, 31 models) and abundance (gray, 10 models). For the effect of a variable to be significantly different from zero, the error bars cannot overlap zero (thin vertical line), e.g., chlorophyll-a had a significant negative effect size for both abundance and species richness, however percent live coral tissue had a significant effect size only in relation to abundance. As a metric of relative model support, the summed model probabilities for the subset of models containing each parameter are listed..... 27

Figure 2.4. Heat map showing the proportion of colonies at a site (see Table 2.S4 for site details) inhabited by each species (see Table 2.2 for species names), calculated for the species observed on 5% or more of colonies (n=751)..... 28

Figure 2.5. For a GLMM with species-specific responses to environmental gradients, sites, and regions (Model 2.S3), the species-specific probability of occurrence trends over environmental factors: (a) colony size, (b) wave height, (c) chlorophyll-a, (d) percent live coral tissue, and (e) depth are plotted with the average trend (across species) in black and data points in gray. See Table 2.2 for species codes. (f) The relative importance of each environmental factor based on average effects (fixed effect estimates) and species-specific responses (standard deviation of random effects)..... 29

Figure 2.6. (a) The first two coordinates of a PCoA on community composition data for species observed on 5% or more of colonies (19 species, 751 colonies). *Trapezia intermedia* (TRIN) is

PREVIEW

the only species in the lower right quadrant. (b) Co-occurrence analysis depicts TRIN as being in 27% of negative co-occurrences (i.e., co-occurrences that were less frequent than expected given a random distribution of species). See Table 2.2 for species codes…….. 30

Figure 2.S1. Paired barplot for percent variation in species richness (a) and community abundance (b) of coral-associated communities (n=708) across spatial scales and environmental factors. Dark gray bars are from a model that included only spatial scales (Model 2.S1). Light gray bars are from a model that also included environmental factors (Model 2.S2)……..….. 37

Figure 2.S2. Mean species richness (a) and community abundance (b) per colony by site, vertical gray lines separating regions. See Table 2.S4 for site coordinates…….. 38

Figure 2.S3. Plots of the average effect (across regions and sites) of environmental factors from GLMMs (Model 2.S2) with species richness (a-h) and community abundance (i-p) as the response variables. Gray bars show the 95% confidence interval on model estimates……..….. 39

Figure 2.S4. Canonical analysis of principal coordinates constrained by sites estimates how much variation in the community composition was explained by variation at the site scale (18%). Sites are shown at their centroids with site codes (see Table 2.S4 for site names)……...….. 40

Figure 2.S5. Variance in occurrences partitioned across spatial scales based on random effect estimates from a spatial GLMM with occurrence as the response variable, survey structure and species-specific spatial patterns included as random effects (a); and an environmental and spatial GLMM with average and species-specific responses to environmental factors added to the components of the spatial model (b). These models correspond to the "Compositional variation over spatial scales" model (a) and the "Full model" (b, Model 2.S3) in Table 2.S2……... 41

Figure 3.1. Within-colony community dissimilarity (gray dots) across a temporal gradient is shown separately for communities at two survey sites. A non-linear model was used to estimate the rate of change, the intercept, and the asymptote for this relationship while allowing variation in the intercept and asymptote across individual colonies nested in sites. The mean site-scale patterns are shown with a solid black line, and colony-scale trends are shown with multi-colored lines for colony-specific ranges along the temporal gradient. The average between colony spatial community dissimilarity for each site is shown with a black dashed line…….....67

Figure 3.2. For each of 61 observed species (Tables 3.S1 & 3.S2), the probability of occurrence on colonies was averaged across surveys (mean spatial occupancy across colonies) and compared

PREVIEW

to the proportion of times the species was observed on an individual colony both at time *t* and at time *t+1* (colony-scale persistence through time). The species data are colored by taxonomic groups and shaped based on their relationship with the host coral. The relationship between spatial occupancy and temporal persistence was modeled with a generalized linear mixed model (black line). A community simulation depicts the expected relationship between spatial occupancy and temporal persistence given random distributions across colonies at each time step (gray line)……...68

Figure 3.3. Original data (Table 3.1) for sea surface temperature (a), significant wave height (b), and surface chlorophyll-a concentration (c) presented in faded colors. Overlaid are summarized versions of these data relative to survey dates (weekly average, daily maximum, and monthly average, respectively). (d) Average alpha diversity, i.e., species richness, and (e) average beta diversity, i.e., Jaccard community dissimilarity distances (e), for each survey date. Data specific to Kewalo are in red, and data specific to Kāneʻohe Bay are in black……..69

Figure 3.S1. Correlations between colony characteristics, i.e., depth, size, average inter-branch distance, and maximum temporal distance between surveys, and colony-specific model estimates for ϕ_1 and ϕ_2. Model estimates are from a nonlinear model of within-colony community dissimilarity over a temporal gradient. ϕ_1 predicts the asymptote for dissimilarity for within-community comparisons over increasingly long periods of time. ϕ_2 predicts the model intercept for dissimilarity within-community comparisons over negligible periods of time……73

Figure 3.S2. Within-colony community dissimilarity (gray dots) across a temporal gradient is shown separately for communities at two survey sites. A non-linear model was used to estimate the rate of change, the intercept, and the asymptote for this relationship while allowing variation across individual colonies nested in sites for the model intercept and rate of change. The mean site-scale patterns are shown with a solid black line, and colony-scale trends are shown with multi-colored lines for colony-specific ranges along the temporal gradient. The average between colony spatial community dissimilarity for each site is shown with a black dashed line. This model had a similar model fit to the best model (ΔAIC = 1.73; Fig. 3.1)……74

Figure 4.1. Inset map of the island of Oʻahu, Hawaiʻi with study site outlined. (a) Location of experimental grid within Kāneʻohe Bay relative to collection site. (b) Arrangement of experimental colonies with treatments blocked by row. Each coral was separated from its closest neighbors by 5 m of rubble, and the entire experimental grid was at least 10 m from natural reef structure……...……...96

PREVIEW

Figure 4.2. The average proportion of corals across treatment groups on which a colonizer type was observed. Output is based on GLMMs that had the proportion of survey windows (considering each coral separately) with occurrence of each colonizer type as the response variable. Models included treatment and colony size (Fig. 4.S2) as fixed effects (Table 4.1), and treatment block (row, Fig. 4.1b) as a random effect……...……...97

Figure 4.3. The number of individuals through time on all corals within each treatment. (a) The number of individuals from Trapeziidae and Alpheidae. (b) The number of commensal and facultative decapod individuals. Species details for xanthid crabs and facultative species are in Table 4.S1……...…….....98

Figure 4.4. (a) Species richness at the coral level and (b) Jaccard dissimilarity at the coral pair level for the colonizing community through time, grouped by treatments. In both plots, points are mean values ± standard errors. Species abundance (Fig. 4.S3) follows a very similar pattern to species richness. (c) The proportion of variation in the composition of the colonizing decapod community explained by treatment group from PERMANOVA analyses run at each time window separately. In each panel, the data points to the right of the vertical dashed line are from surveys of naturally occurring communities around O‘ahu (n=321, surveyed once).……...99

Figure 4.5. Model output for coral growth (percent change per day) from a colonization experiment (coral sizes measured as buoyant weights). Tukey HSD tests were used *post hoc* to identify significant differences between groups. Treatment was significant in the model (p=0.029); groups with the same letter are not significantly different..……...……...100

Figure 4.S1. The natural distribution of *A. lottini* and *Tr. intermedia* from surveys of 321 *Pocillopora meandrina* coral colonies around O‘ahu (within the size range of experimental corals and with at least 50% live tissue, Counsell et al. 2018). (a) The proportion of corals with 0, 1, 2, or 3 *A. lottini*; (b) with 0, 1, 2, or 3 *Tr. intermedia*; and (c) with *A. lottini* only, *Tr. intermedia* only, neither species, or both species. Colored intervals represent the null distributions of occurrence if individual *A. lottini* and *Tr. intermedia* were distributed independently of other *A. lottini* and *Tr. intermedia* individuals.……102

Figure 4.S2. Plots of the average proportion of corals across a gradient of coral colony sizes for which a colonizer type was observed from GLMMs. Each model included treatment (Fig. 4.2) and colony size as fixed effects (Table 4.1), and treatment block as a random effect. Gray bars show 95% confidence intervals on model estimates.……...103

PREVIEW

Figure 4.S3. Mean (± standard error) colony level species abundance during a community establishment experiment for four treatments. Patterns are very similar to those of species richness (Fig. 4.4a). The data points to the right of the vertical dashed line are from surveys of naturally occurring communities.……....104

Figure 4.S4. NMDS plots (k=2; stress=0.1846) through time for the decapod community on corals manipulated at month 0 to have a pair of *A. lottini*, a pair of *Tr. intermedia*, a pair of both, or neither (control treatment). A NMDS plot (k=2; stress=0.2209) based on surveys of natural communities (n=321) separated based on natural presence of *A. lottini* and/or *Tr. intermedia*. In all plots, community composition is shown as axis scores from non-metric multidimensional scaling (function "metaMDS" in *R* package *vegan*; Oksanen et al. 2017).....105

Figure 4.S5. Model output for coral growth (percent change per day) from an establishment experiment (coral sizes estimated from photographs taken at month 0 and month 6). At month 0, corals were manipulated to have a pair of *A. lottini*, a pair of *Tr. intermedia*, a pair of both, or neither (control treatment), and then communities were followed through time as other individuals colonized and became established on the corals. Treatment did not have a significant effect on coral growth (p=0.148).……..……..106

PREVIEW

CHAPTER 1

Introduction

Community assembly

An ecological community is an assemblage of species that live within a shared environment and are connected through a network of potential and realized influences on each other. Understanding the community assembly dynamics (Diamond 1975) that drive the formation and maintenance of communities over various spatial and temporal scales is one of the fundamental goals of ecology. Which subset of species arrives to a community (i.e., community formation) hinges on successful dispersal or emigration (Fox et al. 2012; Basterretxea et al. 2013; Vaz et al. 2013) and favorable environmental characteristics (Belyea and Lancaster 1999; Thuiller et al. 2015). In addition, community formation is often affected by the order in which species arrive, through positive and negative, direct and indirect species interactions (Sutherland 1974; Connell and Slatyer 1977; Booth 1992). After species arrive to a community, the successful retention of species (i.e., community maintenance) depends on their fitness and survival in the community. These species-specific metrics are influenced by habitat quality (Hixon and Beets 1993; Macneil et al. 2009; Almeida et al. 2014) and species interactions (Connell 1961; Stallings 2008; Stier and Leray 2014). The structuring processes that operate on the formation and maintenance components of community assembly lead to observable patterns in the composition of communities, which can in turn be used to provide support for potential underlying processes (Schmitt and Holbrook 1999; Depczynski and Bellwood 2005; Paradise et al. 2008; Sim-Smith et al. 2013; Jankowski et al. 2015).

To investigate patterns in ecological communities and to illuminate relative support for potential mechanisms structuring communities, a combination of analyses can be used that consider variation in (1) single species occurrences, (2) community-level metrics (e.g., abundance of individuals, biomass, species richness, species evenness), and (3) multivariate community composition data. Analyses of individual species can depict important responses to environmental drivers, and some species can be utilized as indicator species for patterns in the broader community (King and Beazley 2005). Univariate community-level metrics of biomass

PREVIEW

and species diversity can be very effective for identifying patterns in community composition, including shifts in species diversity over large spatial scales (Boltovskoy and Correa 2017), peaks in diversity relative to disturbances (Sousa 1979; Johst and Huth 2005; England et al. 2008), and associations between biomass and environmental drivers (Heenan et al. 2016). Species turnover, a quantification of the compositional difference between communities (Sørensen 1948), can provide valuable insight for understanding processes that lead to differences in community composition without leading to differences in the number of species (i.e., interchange of specific species between sites). Species turnover can be considered as a metric of diversity with which to compare the variability of communities between different groups of communities (Whittaker 1960; Martins et al. 2018). Species turnover can also be used to compare the composition of communities along a directional gradient relative to environmental characteristics (Silva et al. 2018), space, or time (Anderson et al. 2011). To directly investigate patterns in multivariate community composition data, ordination analyses can be utilized. These types of analyses include principal components analyses (PCA) and non-metric multi dimensional scaling (NMDS), both of which enable simultaneous consideration of numerous species observed across multiple communities. In addition, recent shifts to open-source statistical software packages have increased the use of hierarchical mixed models and non-linear models to evaluate complex patterns in communities over multiple scales (Bates et al. 2015; Gove et al. 2015; Pinheiro et al. 2016; R Development Core Team 2016).

Observational studies of naturally occurring communities are a core component of ecology. Surveys can often be conducted with minimal impact to the natural community. Basic patterns in compositional variation can be characterized. Patterns in variation can be considered over large spatial and temporal scales, and along broad gradients in environmental drivers (Freestone and Inouye 2015). However, inferences of underlying processes based on observed patterns should be approached with caution (Connor and Simberloff 1979; Wilson 1994; Fox 2013; Barner et al. 2018). The patterns expected for a given process may not be realized in communities that are exhibiting nonequilibrium dynamics (Loreau 2000; Hixon 2011). In addition, there is potential for complex interactions across multiple underlying processes in which the overall outcome does not reflect a linear combination of processes (Hixon and Carr 1997; Barner et al. 2018). Further, neutral theory and null models can explain some of the

PREVIEW

patterns observed in communities (Connor and Simberloff 1979; Chave 2004; Hubbell 2005). These inherent complexities in the interpretation of community composition patterns should not deter observational studies. Instead, an awareness of these intricacies should encourage the cautious interpretation of observed patterns. Investigations into the mechanisms structuring communities can be strengthened through combinations of spatial and temporal surveys (Freestone and Inouye 2015). Moreover, observational studies can be paired with experiments that directly test for the effects of potential underlying mechanisms (Stier and Osenberg 2010; Miller and TerHorst 2012; Stier et al. 2013; Martins et al. 2018).

Study system

This research focuses on the semi-cryptic coral reef community associated with the scleractinian coral *Pocillopora meandrina* Dana, 1846. Pocilloporid corals provide an ideal habitat for studying community assembly dynamics because they: (1) form small, spatially discrete habitat patches, (2) are broadly distributed across much of the Indo-Pacific, and (3) host a diverse suite of fish and invertebrate species (Stella et al. 2010). In Hawai‘i, where this research takes place, *P. meandrina* is one of the most common corals, and in many areas, it provides the only branching reef structure.

The community associated with *Pocillopora* corals includes a collection of species that have been shown to have a mutualistic relationship with the host coral. Trapeziid crabs defend their *Pocillopora* host from the predatory crown-of-thorns sea star (Pratchett 2001; McKeon et al. 2012) and from the deleterious effects of vermetid epibionts (Stier et al. 2010). In addition to providing protective services to their host coral, trapeziid crabs can increase coral growth and survival by removing sediments from the coral tissue (Stewart et al. 2006). Similarly, snapping shrimp remove sediments from the coral tissue (Stier et al. 2012) and help protect their host coral against corallivores (Pratchett 2001; McKeon et al. 2012). Damselfishes sheltering within the branches of *Pocillopora* colonies can also promote growth and survival of the host coral through multiple mechanisms. The territorial behaviors of damselfish can minimize the predation of coral polyps by other reef fishes (Gochfeld 2009; Chase et al. 2014). Damselfish that exhibit a sleep-swimming behavior circulate the water between coral branches at night (Goldshmid et al. 2004). In addition, damselfishes that shelter within the coral’s branches provide the host coral with

PREVIEW

excretion-based nutrient enrichment (Holbrook et al. 2008). These mutualistic species and many other species that associate with *Pocillopora* corals can be effectively surveyed with non-destructive survey techniques (Sin and Lee 2000).

Research goals

To study spatial scales of variation, sets of communities were surveyed at multiple sites spanning the entire coastline of the island of O‘ahu, Hawai‘i (Chapter Two). This combination of a high replication of surveys at small spatial scales and a large spatial distribution of survey sites enabled the use of hierarchical mixed models to compare variation in community composition with environmental gradients at both the colony- and site-scale. In addition to examining patterns in community level parameters, multivariate hierarchical mixed models were developed to depict species-specific patterns. Further, the co-occurrence of commonly observed species was analyzed to identify non-random patterns of habitat use relative to the distribution of other species.

To elucidate the consistency of these communities through time, focal communities were followed quasi-monthly for over three years (Chapter Three). Within-colony variation in community composition was analyzed relative to a temporal gradient and compared to between colony variation. The persistence of each species on individual host colonies was compared to their spatial occupancy across colonies. Mantel tests and linear regressions were used to investigate potential correlations between temporal variation in communities and synchronous fluctuations in regional environmental parameters. Transition probability models were developed to estimate the effect of functionally similar species on arrival and departure dynamics.

To directly test for ‘priority effects’, a potential mechanism of community assembly, colonies were arranged in an experimental grid and their associated communities were manipulated to create distinct community treatments (Chapter Four). The treatments focused on two commonly observed, protection mutualist species and followed a factorial design resulting in the following four treatment groups: (1) a pair of alpheid shrimps, (2) a pair of trapeziid crabs, (3) pairs of both alpheid shrimps and trapeziid crabs, or (4) an empty coral (control group). The experiment was divided into two sections to separately study how the presence of established protection mutualist species affected community formation and community maintenance. First,

PREVIEW

the arrival of species to corals was recorded, and probabilities of species colonization were compared among treatments. Then, community composition was followed through time, and the trajectories of functional groups, as well as the trajectories of metrics characterizing community composition, were compared among treatments.

Overall, this dissertation characterizes the reef communities associated with *P. meandrina* in Hawai‘i. Patterns in the natural variation of these communities are explored over spatial and temporal scales. Probable associations between these patterns and environmental gradients at the colony- and site-scales are evaluated. Evidence for species-interactions as a potential mechanism structuring these communities is considered from species-specific patterns recorded through spatial and temporal surveys. To directly examine the role of species interactions, a community manipulation tests the effect of two species on community assembly processes. This dissertation’s combination of observational and experimental research provides a template of analyses for investigating spatial and temporal patterns in ecological communities.

PREVIEW